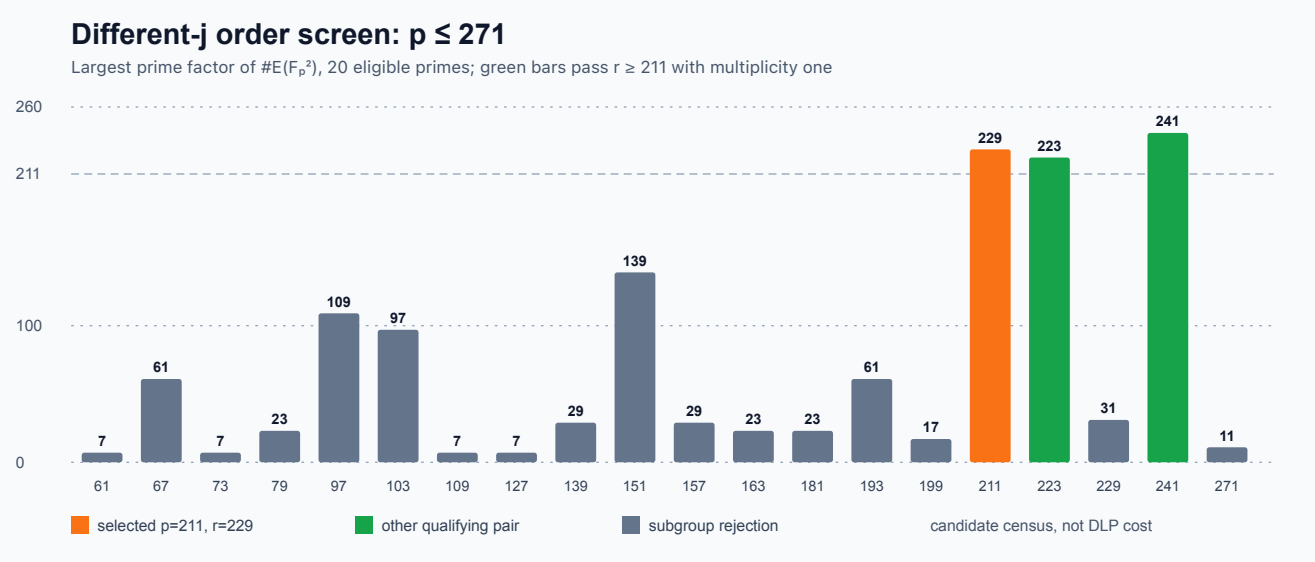
Bounded larger-order different-j isogeny screen

2026-10-15 · native Rust · Darwin arm64, macOS 26.6 · public synthetic curves. The protocol and exact source/binary hashes were committed before execution. All 20 eligible primes with $61 \le p \le 271$ and $p = 1 \pmod 3$ were emitted. Seventeen failed the frozen subgroup criterion, and three passed. The first passing p was selected as specified.

p	Both orders over $F(p^2)$	r	Cofactor	Source j	Dest. j
211	**44,655**	**229**	**195**	**77**	**160**
223	50,175	223	225	183	222
241	58,563	241	243	33	164

The selected source is $y^2=x^3-21x+37$ with kernel generator (3,1); the explicit degree-three destination is $y^2=x^3-81x-243$. Its 44,655 points factor as $3\cdot5\cdot13\cdot229$. Independent field and point formulas checked **192/192** sampled map witnesses. A separately frozen direct $F(p^2)$ quadratic-character census counted both curves at each accepted p and matched all three trace-derived orders. Different j values exclude an isomorphism between the two selected models.



The graph is a candidate census, not a DLP cost plot. It shows the largest prime factor of each group order and the $r \ge 211$ selection threshold. The selected order-229 subgroup has 228 distinct nonidentity points, enough for the planned 64-target batch. A subsequent separately frozen full pointwise map/fiber audit passed; see FIBER_REPORT.pdf. Relation collection, scalar recovery, direct BSGS/rho comparison, counted S, isolation and independent-host performance replay remain pending.

Evidence: RAW_V1.jsonl SHA-256 31238914df366380; REPLAY_V1.json ac2c13d5496755d5; CENSUS_V1.jsonl 50f2b79d9811c72c. The source-linked RESULTS.md records full hashes, scripts, empty stderr receipts, checker scope, and the next test. This screen does not change the canonical ECDLP frontier.